Escaping the Echo: Momentum-Coupled Phase Emission for Generative Resonance in Synthetic Intelligence

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Abstract In wave-based cognitive substrates utilizing Continuous Morphological Dynamics (CMD), external stimuli inherently generate maximal amplitude at the point of injection, leading to "Stimulus Reflection" (the parrot effect) during syntax emission. We propose a novel non-algorithmic architecture, **Momentum-Coupled Phase Emission (MCPE)**, which transitions the readout mechanism from absolute amplitude to kinetic wave momentum. By coupling syntax oscillators exclusively to the first derivative of the energy field ($dA/dt > 0$), the substrate naturally ignores decaying direct stimuli and phase-locks exclusively to emergent, forward-scattering internal thoughts.

1. Introduction: The Acoustic Blinding Problem

In the current iteration of the SI engine, the Kuramoto syntax slots (Subject, Verb, Object) phase-lock to cavities with the highest absolute amplitude (A). When external input (e.g., "notch created minecraft") drives the field, those specific cavities reach maximum A. Before the wave can propagate deep into the latent graph to retrieve a meaningful association, the syntax slots lock onto the massive input wave, resulting in a mirrored output. The system is structurally blind to its own internal resonance because the external drive acts as an acoustic flashbang.

2. Novel Architecture: Momentum-Coupled Phase Emission (MCPE)

To solve this without algorithmic filtering, we introduce a Hamiltonian shift in the readout layer. Instead of coupling syntax slots to the "position" (Amplitude, A) of the wave, we couple them to the "momentum" (Flux, P).

2.1. First-Derivative Readout (dA/dt Gating)

In a physical wave system, a cavity directly hit by a stimulus reaches peak energy and immediately begins to disperse it (cooling down). Its amplitude is high, but its momentum is negative ($dA/dt \leq 0$). Conversely, deep-graph nodes that are currently being activated by internal resonance are absorbing energy; their amplitude might be lower, but their momentum is strictly positive ($dA/dt > 0$).

We redefine the coupling strength K_{emit} between a cavity i and a syntax slot as:

$$K_{emit} = \Theta\left(\frac{dA_i}{dt}\right) \cdot A_i$$

Where $\Theta(x)$ is the Heaviside step function (returns x if $x > 0$, else 0). *Result:* Directly driven words (the user's prompt) instantly decouple from the output slots the millisecond the prompt ends, because they are decaying. The output slots are physically forced to lock onto the *ripples* (the internal thoughts), not the *splash* (the user's input).

2.2. Thermodynamic Refractoriness (The "Acoustic Mass")

To prevent the wave from bouncing directly back to the input nodes, we modify the cavity inertia. When a cavity is driven externally, it accumulates temporary "acoustic mass" M_i:

$$\frac{dM_i}{dt} = \gamma I_{ext} - \delta M_i$$

Higher mass lowers the cavity's fundamental oscillating frequency temporarily, pushing it out of the L6-A (3Hz) emission band. It retains its associative links but becomes too "heavy" to be picked up by the Kuramoto slots.

2.3. Temporal Settle Separation (Perception vs. Articulation)

We separate the physics tick into two distinct phases using a global damping coefficient ζ:

1. **Injection Phase (T_0 to T_1):** External wave hits. ζ is high. Energy is absorbed. Syntax slots are decoupled ($K = 0$).
2. **Articulation Phase (T_1 to T_2):** External drive is 0. ζ drops. The wave propagates through the L6-A bridges. Syntax slots engage, seeking positive momentum ($dA/dt > 0$).

3. Expected Dynamics

Scenario: User inputs `insulin controls glucose`.

1. T_0 **(Injection):** Cavities `insulin`, `controls`, and `glucose` spike in absolute amplitude ($A \approx 1.0$). Their mass M_i increases.
2. T_1 **(Propagation):** Input ceases. The energy from `insulin` and `glucose` propagates through the ontology graph to connected cavities like `metabolism`, `pancreas`, and `glycogen`.
3. T_2 **(Articulation):**

- `insulin` has high A, but $dA/dt < 0$ (cooling). It is ignored.
- `pancreas` has medium A, but $dA/dt > 0$ (warming up). It is locked.
- The Kuramoto slots phase-lock to `pancreas regulates metabolism` rather than parroting the input.

Conclusion: MCPE preserves the zero-rule continuous dynamics of the substrate while physically guaranteeing that the engine generates consequences rather than echoing stimuli.
